Team Web Application Project Specification

Loan Application Decision Workflow

Project Description

This project is a web application designed to make the in-branch loan application and approval process faster and easier for financial institutions.

Many financial institutions have made their online loan applications quick and modern, but the process used by employees inside a branch can still feel slow and outdated. Loan officers may have to move between different systems, collect information manually, pull credit, write notes, and then send information to a credit manager for a decision.

The goal of this application is to create a simple internal system that allows loan officers to quickly create an application, collect the important applicant and credit information, and send the application to a credit manager. Credit managers can then review applications from a queue and approve or deny them.

The first version will focus on making this core process simple rather than trying to handle every possible part of loan underwriting.

Target Users

Loan Officers

- Create a new loan application.
- Enter applicant information.
- Enter income and employment information.
- Enter basic housing information.
- Enter the requested loan amount and purpose.
- Record credit information.
- Record debts and monthly payments.
- Review basic ratios or other useful financial information.
- Write notes explaining the application.
- Submit the application to a credit manager for review.
- View the status of applications they have submitted.

Credit Managers

- View a queue of applications waiting for a decision.
- Open an application and review all applicant information.
- Review credit information and basic ratios.
- Review the loan officer's notes.

- Add their own notes.
- Approve an application.
- Deny an application.

Administrators

- Add users.
- Edit users.
- Delete or deactivate users.
- Assign roles.
- View users and their current roles.

MVP Features

User Authentication

Users should be able to sign in and access features based on their assigned role.

Dashboard

Users should see a dashboard when they log in.

The dashboard may show applications recently created, waiting for review, approved, or denied. The information shown may depend on the user's role.

Create Application

Loan officers should be able to create a new application for one applicant.

The application should collect first and last name, contact information, address, employment information, income, housing status, monthly housing payment, requested loan amount, and loan purpose.

Credit Information

The application should store enough basic credit information to help the credit manager make a decision.

This may include credit score, total monthly debt payments, income, housing payment, basic financial ratios, other important credit information, and loan officer notes.

The goal is not to create a complete automated underwriting system. The information should simply help the credit manager understand the applicant's financial situation.

Submit Application

After entering the required information, the loan officer should be able to submit the application. Once submitted, the application should appear in the credit manager's review queue.

Credit Manager Queue

Credit managers should see a list of applications that are waiting for a decision.

The queue should show basic information such as applicant name, requested loan amount, loan officer, credit score, application status, and date submitted.

Review Application

A credit manager should be able to open an application and review all of the information entered by the loan officer.

The credit manager should be able to review applicant information, employment and income information, credit information, financial ratios, loan officer notes, add notes, and approve or deny the application.

Application Status

Each application should have a simple status: Draft, Submitted, Approved, or Denied.

Loan officers should be able to see the current status of their applications.

User Management

Administrators should have a user management screen where they can add users, edit users, delete or deactivate users, and assign users as loan officers, credit managers, or administrators.

Optional or Future Features

- Multiple applicants or co-borrowers.
- Automated credit report integrations.
- More advanced debt-to-income calculations.
- Automated approval rules.
- Different loan products.
- Mortgage applications.
- Business loans.
- Collateral information.
- Document uploads.
- Electronic signatures.
- Loan conditions.
- Counteroffers.
- Sending applications back to loan officers for additional information.
- Notifications.
- Application history and auditing.
- Funding and closing workflows.
- Reporting and analytics.
- Customer-facing online applications.

Basic User Flows

Loan Officer Flow

- Sign in and open the dashboard.
- Select New Application.
- Enter applicant, employment, income, credit, and debt information.
- Review useful ratios or financial information.
- Add notes about the applicant.
- Review and submit the application.
- Application status changes to Submitted and appears in the credit manager queue.
- Later view whether the application was approved or denied.

Credit Manager Flow

- Sign in and view the application queue.
- Select an application.
- Review applicant information, income, debt, credit score, ratios, and loan officer notes.
- Add notes if needed.
- Approve or deny the application.
- Application status changes and the loan officer can see the decision.

Administrator Flow

- Sign in and open User Management.
- View current users.
- Add a new user and assign a role.
- Edit, deactivate, or delete existing users.

Key Screens

- Login
- Dashboard
- New Application
- Application Details
- Credit Manager Queue
- Credit Decision
- User Management

Initial Data Models

User

- User ID
- First name
- Last name

- Email
- Role
- Active status

Application

- Application ID
- Applicant ID
- Loan officer ID
- Requested loan amount
- Loan purpose
- Application status
- Date created
- Date submitted
- Decision date

Applicant

- Applicant ID
- First name
- Last name
- Phone number
- Email
- Address
- Housing status
- Monthly housing payment

Employment and Income

- Employment ID
- Applicant ID
- Employer name
- Job title
- Employment status
- Monthly or annual income

Credit Information

Credit information ID

Application ID

Credit score

Monthly debt payments

Income used for ratios

Housing payment

Calculated ratios

Application Notes

- Note ID
- Application ID
- User ID
- Note text
- Date created

Decision

- Decision ID
- Application ID
- Credit manager ID
- Decision
- Decision notes
- Decision date

Important Application Interactions

- Sign in.
- Get the current user's information and role.
- Create an application.
- Save or update an application.
- Add applicant information.
- Add income and employment information.
- Add credit information.
- Calculate basic ratios.
- Submit an application.
- Retrieve applications created by a loan officer.
- Retrieve applications waiting in the credit manager queue.
- Retrieve a specific application.
- Add notes to an application.
- Approve an application.
- Deny an application.
- Create a user.
- Edit a user.
- Assign a role.
- Delete or deactivate a user.

Possible API Actions

- Get current user
- Get users

- Create user
- Update user
- Delete or deactivate user
- Get applications
- Create application
- Get application details
- Update application
- Submit application
- Get submitted applications
- Add application note
- Approve application
- Deny application

Out of Scope for the First Version

- Multiple applicants on one loan.
- Every type of consumer or commercial loan.
- Automatic credit decisions.
- A complete underwriting system.
- Real credit bureau integrations.
- Document signing.
- Loan funding.
- Complex collateral.
- Advanced underwriting calculations.
- Complicated approval conditions.

Open Questions

- Which exact financial ratios should be shown to credit managers?
- How much debt information should the loan officer enter?
- Should loan officers be able to edit an application after submitting it?
- Should a denied application be editable and resubmitted?
- Should deleted users actually be deleted or simply deactivated?
- What information should appear on the dashboard?
- What fields should be required before an application can be submitted?
- Should credit manager notes be visible to loan officers?
- Should applications have an activity history?